

ORION 1.0

Problem Statement ID: ORION-PS-03

Problem Statement Title: SylvaSense

Team ID: [FILL THIS IN]

Team Name: Order of One

Live demo: [PASTE YOUR URL]



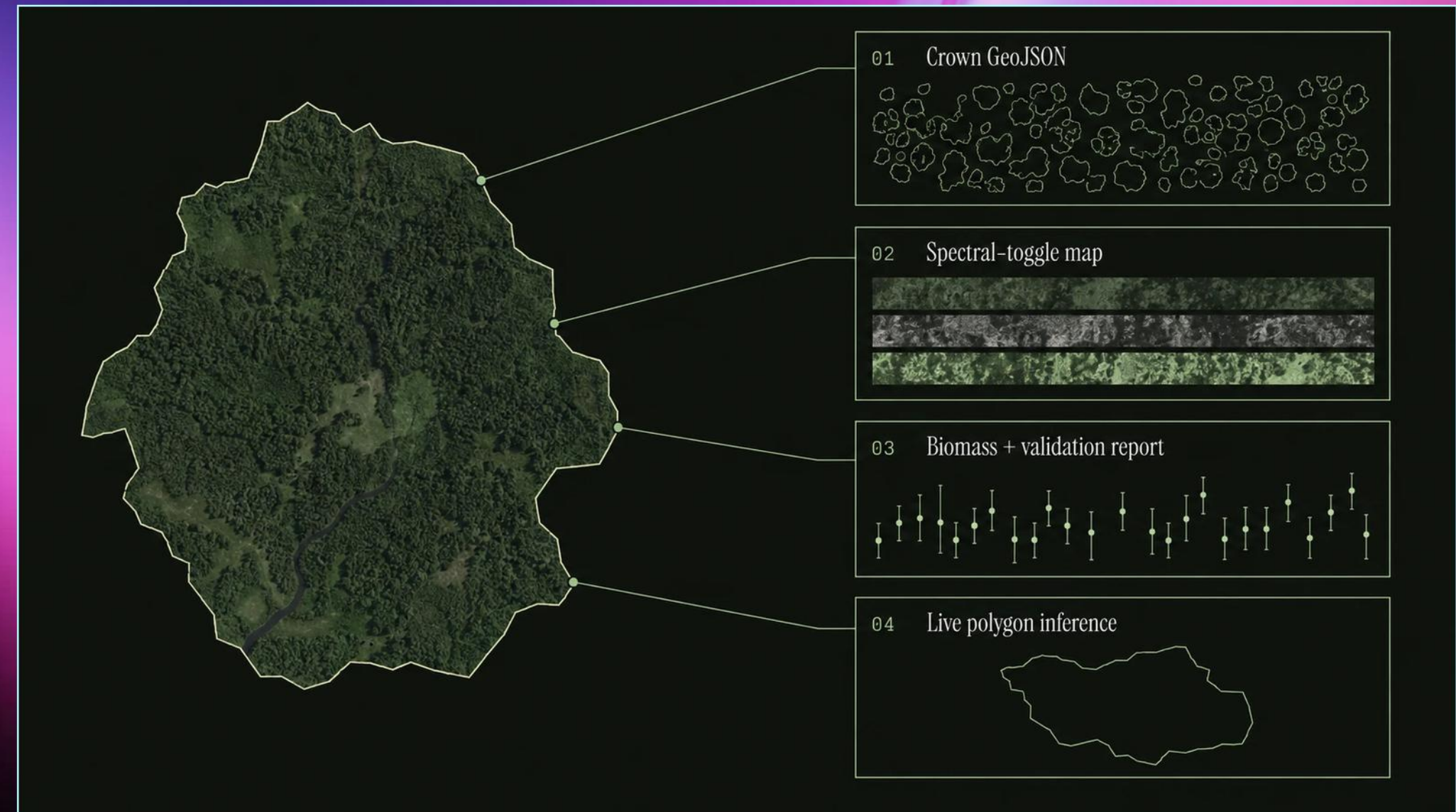
Proposed Solution

SYLVASENSE

From canopy to carbon.

One polygon in. Four kinds of evidence out.
Every number carries its uncertainty.

- 01 Crown GeoJSON**
Per-tree polygons, each with a confidence score
- 02 Spectral-toggle map**
Optical, radar and structure in one frame
- 03 Biomass + validation report**
Mg/ha with an interval that is checked
- 04 Live polygon inference**
Draw an area; evidence returns with its sources





Technical Approach

Three scales. One pipeline.

CROWNS / FINE AERIAL IMAGERY

Mask2Former + Swin-S

0.1 m NEON RGB, tiled and merged

Every mask checked against the canopy height model

BIOMASS / OPTICAL + SAR + LIDAR

Gradient boosting + split conformal

GEDI footprints are the calibration target

Coverage measured on held-out ground, not assumed

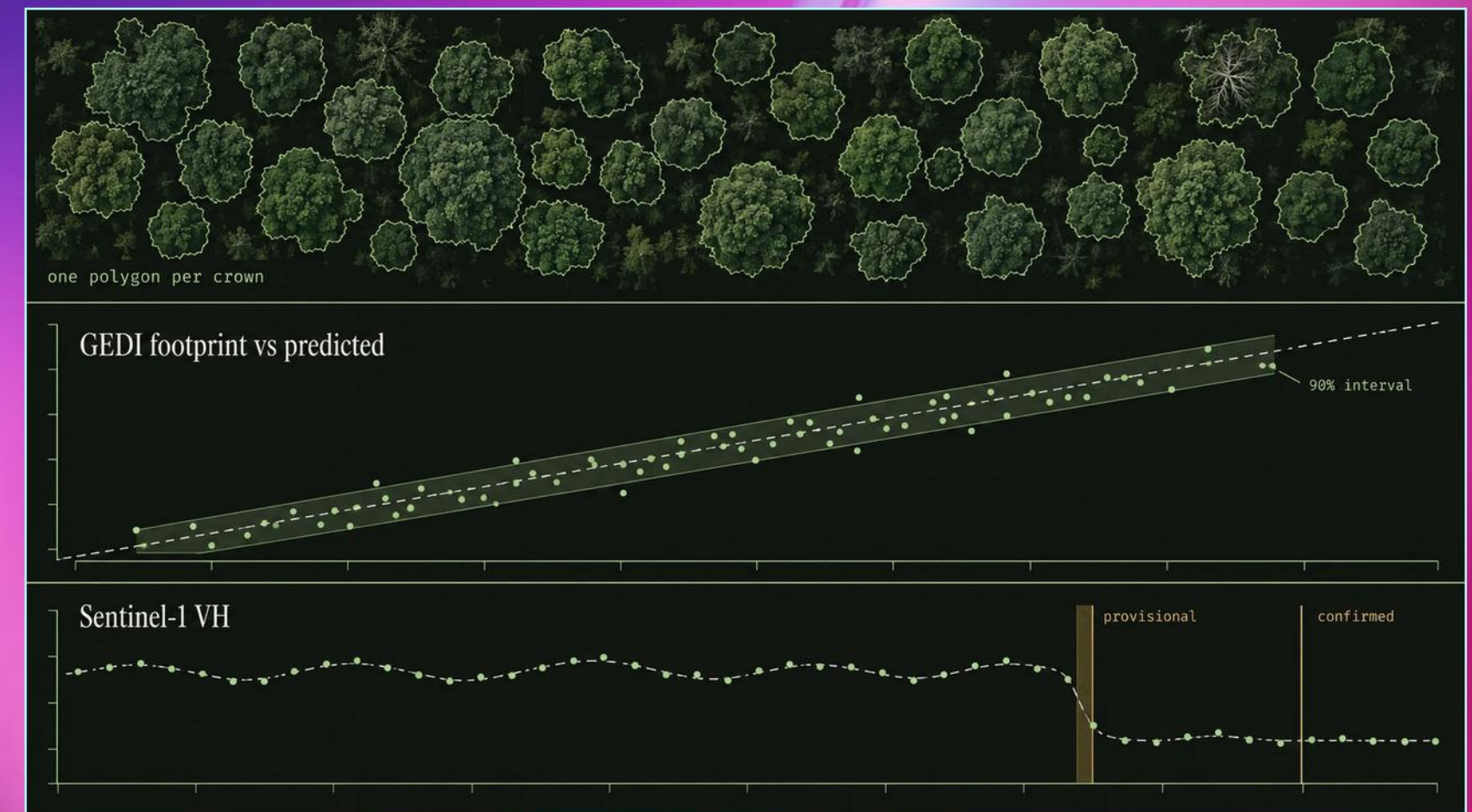
CHANGE / MULTI-YEAR OBSERVATIONS

Harmonic baseline + Bayesian confirmation

A drop must persist before it is called a detection

Below its power threshold the detector reports nothing

AGB = Σ density $\times$ area | **Aboveground carbon = AGB $\times$ CF**

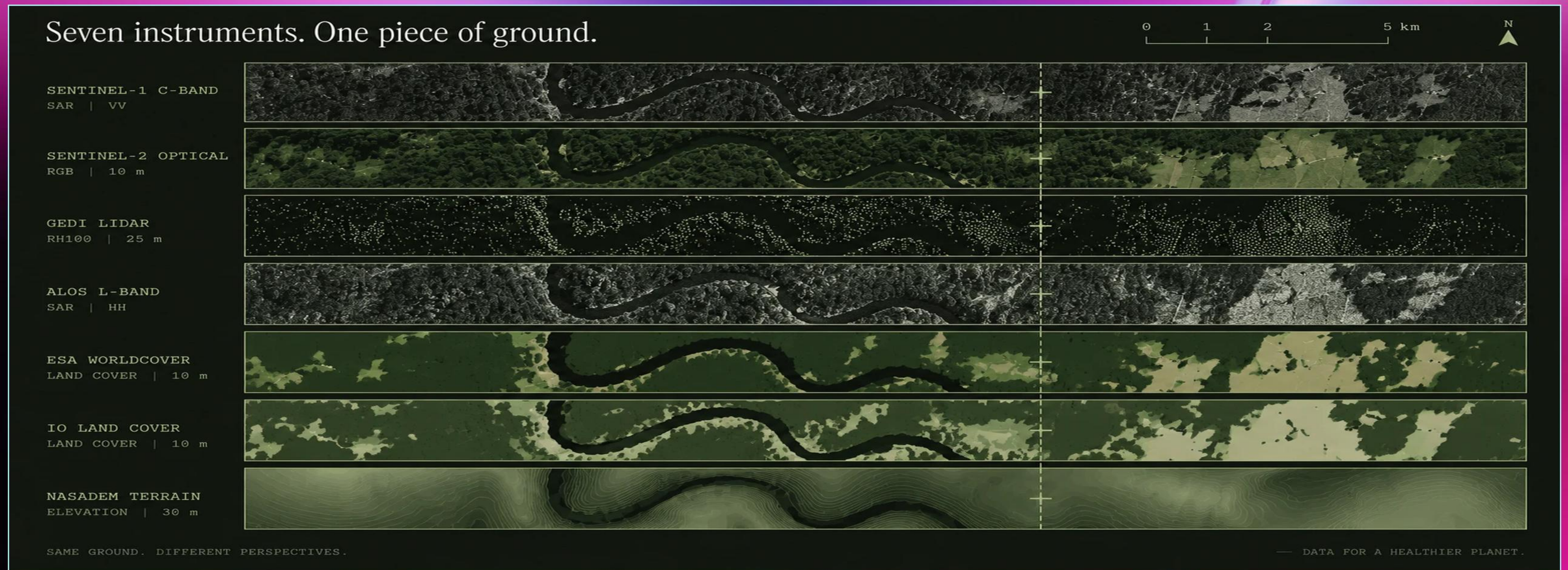




Convergence

No single sensor is decisive.

C-band and L-band radar, optical, lidar and three land-cover products.
Their errors are uncorrelated, which is what makes agreement mean anything.





Feasibility & Viability

Three gates. Already passing.

01 VERIFIED THE DATA

6 sites audited on 3 independent layers

2 of 5 candidates failed the audit and were rejected

02 EARNED THE MODEL CHOICE

Spatially-blocked folds, never a random split

Neighbouring footprints are one observation, not two

03 MEASURED THE TRADE-OFFS

False positives on primary forest: 0.01–0.02%

Coverage holds on unseen blocks — and one site fails, visibly

41.6% → 0.01%

A 365-day window was confirming a national park as cleared. Persistence is a question about time. The gate is now a span.

Measured on 6 sites, 2024–2026 • 60 automated tests • Rondônia, Brazil

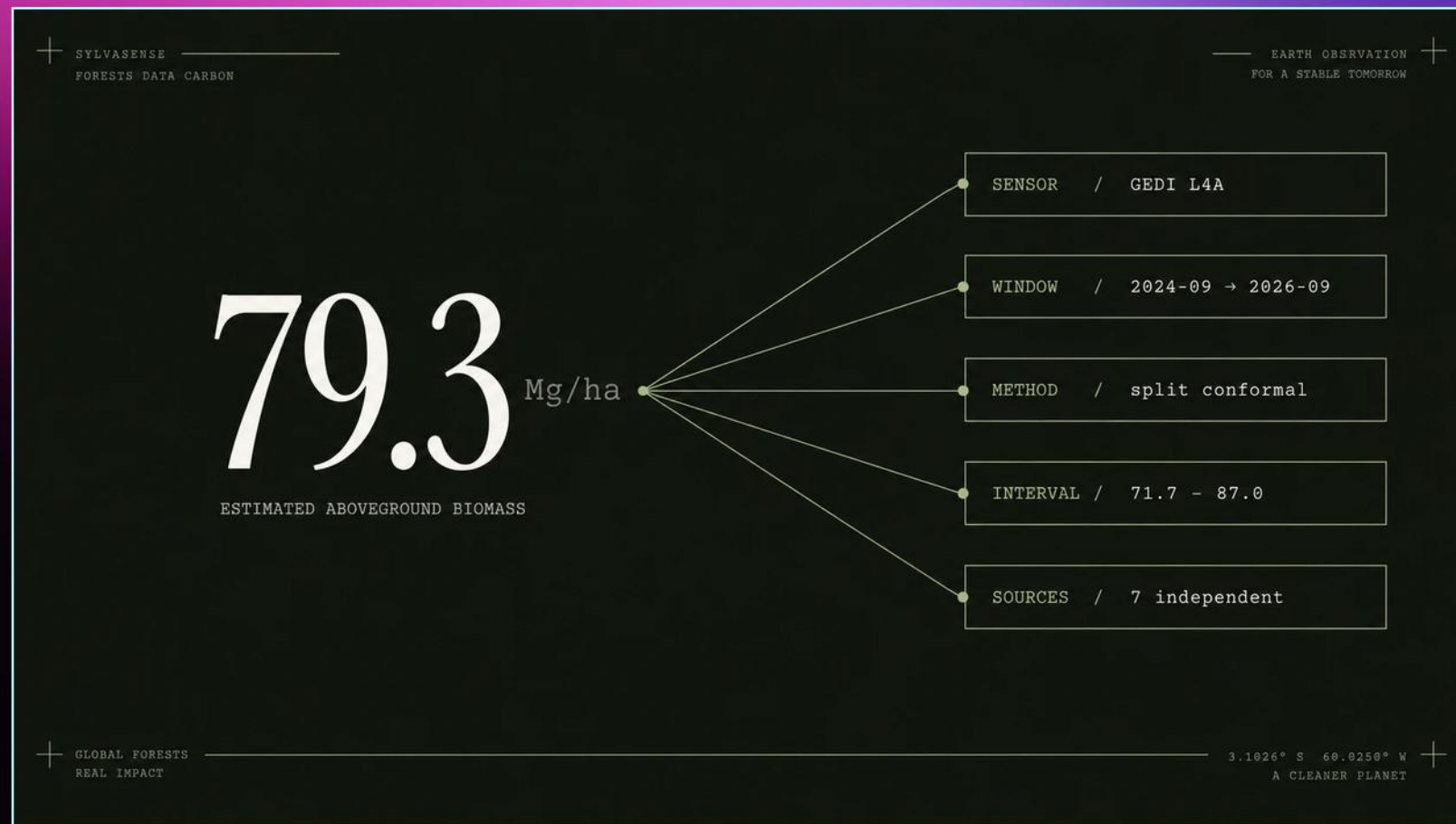




Confidence

Every number carries its source.

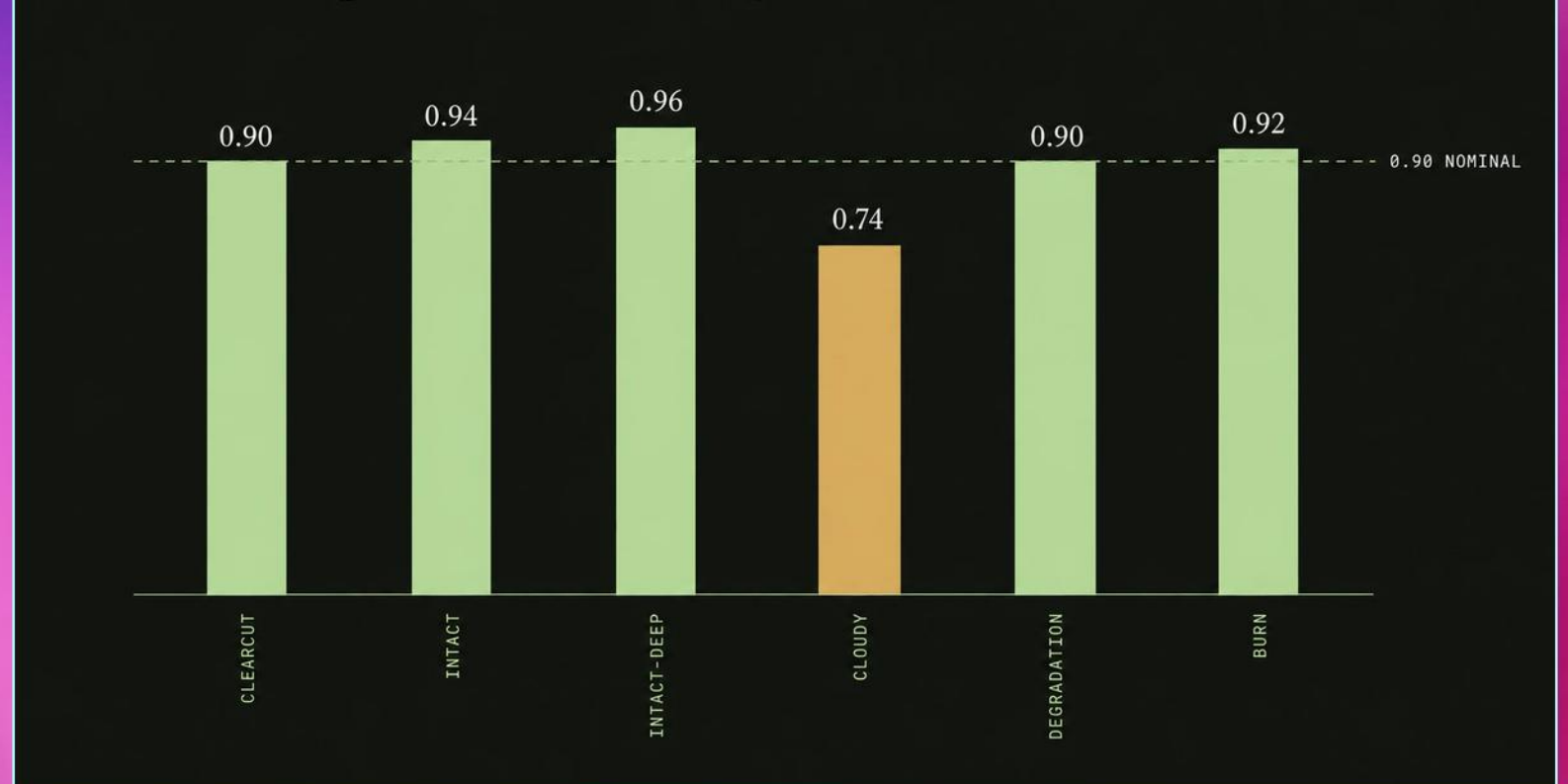
And the interval on it is measured on ground the model never saw — including where it fails.



One value, everything needed to audit it

Sensor, window, method, interval and the count of independent sources travel with the number itself.

Interval coverage, measured on unseen ground.



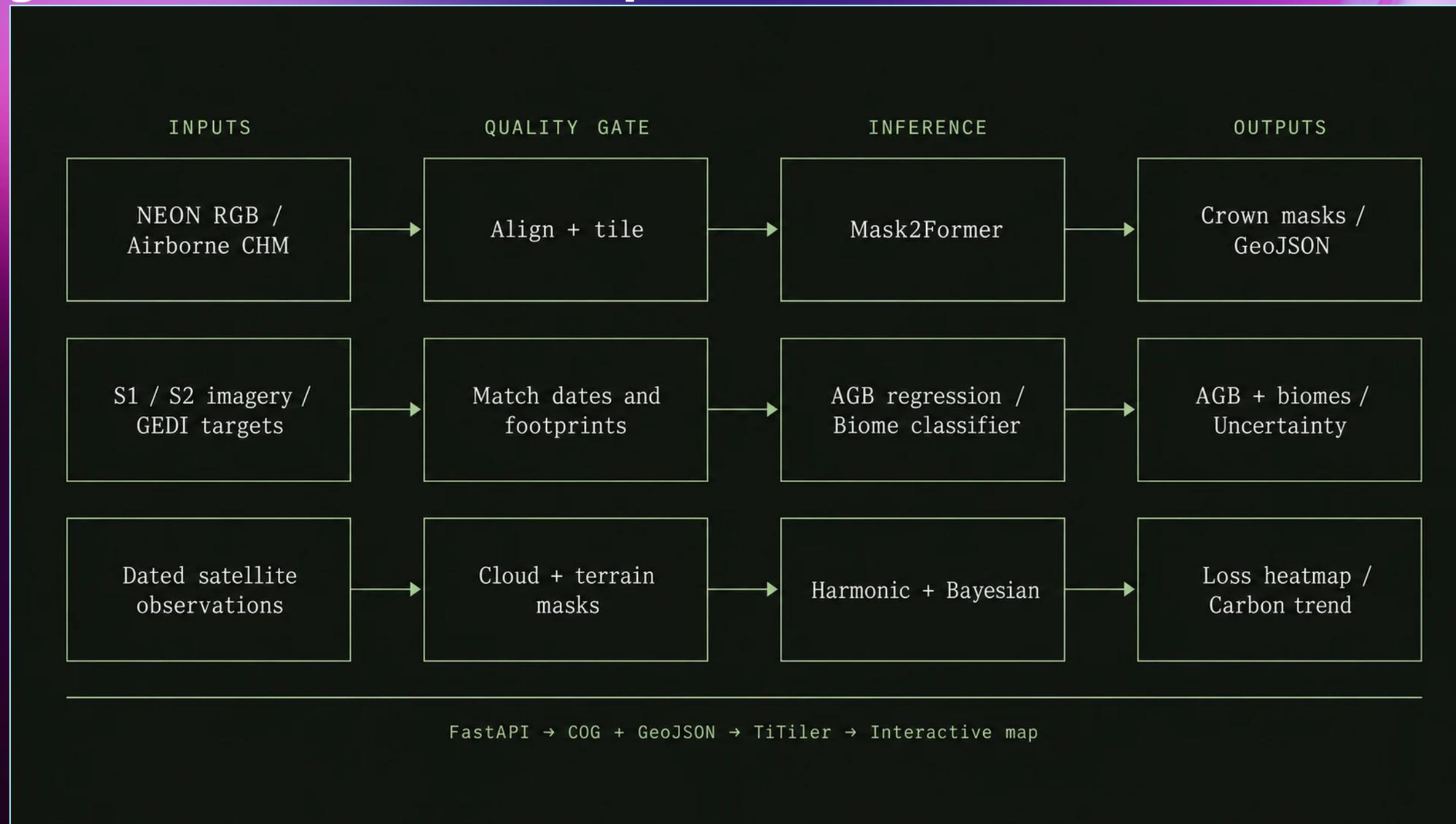
Five of six sites pass

The wet-season interfluvium reaches 0.74 against a nominal 0.90, and the product says so rather than hiding it.



Architecture Diagram

One polygon. Three coordinated paths.





Tech stack

DATA

Earth Engine · Planetary Computer · GEDI

PREPARE

Rasterio · GDAL · GeoPandas · odc-stac

MODEL

PyTorch · Mask2Former · split conformal

SERVE

FastAPI · Redis / Celery · Docker

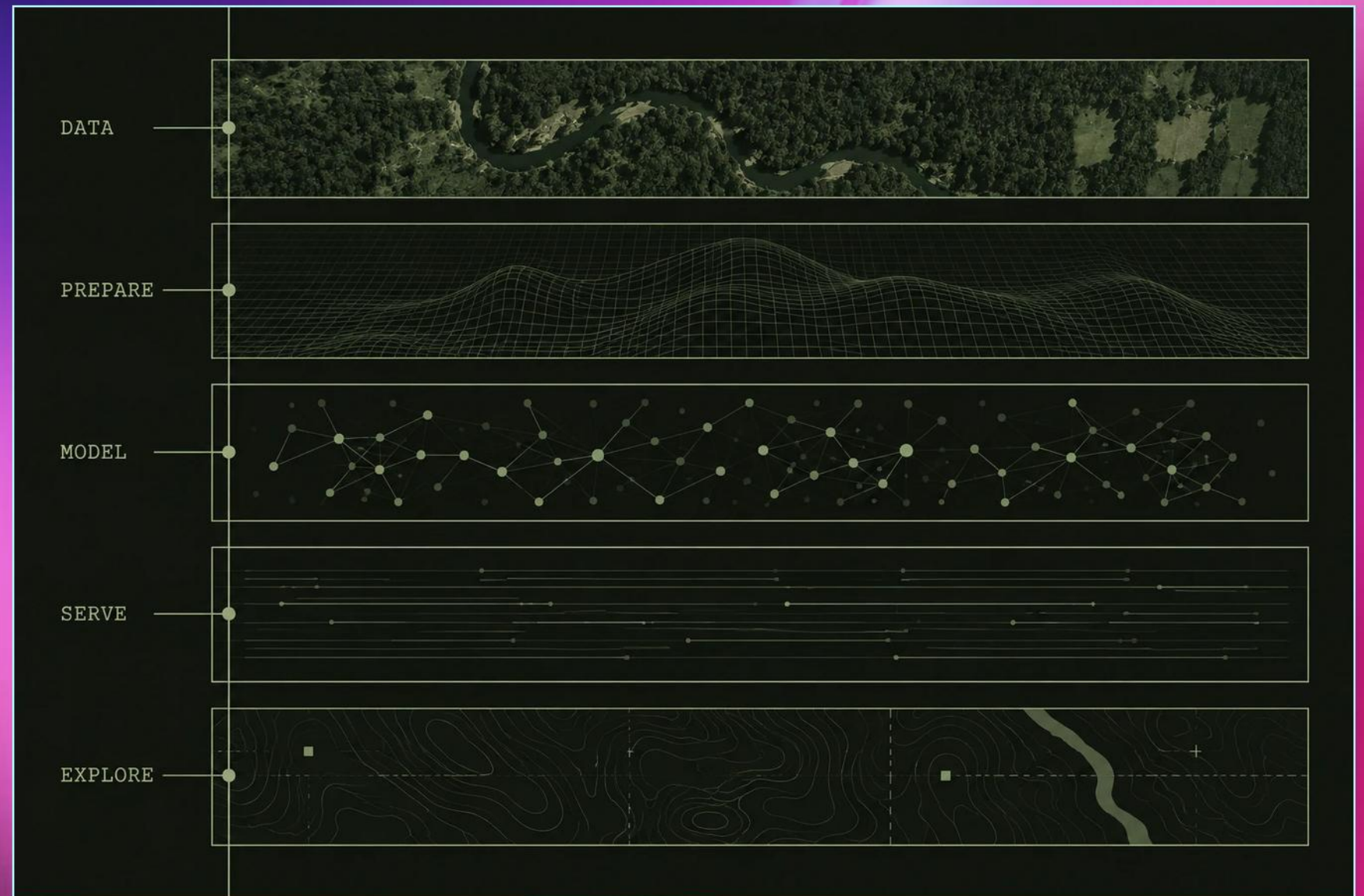
EXPLORE

TiTiler · Mapbox GL JS · deck.gl

Windowed COG reads. Cached reports.

Cold report 151–222 s · cached 3–23 ms

7 sources, each free to fail alone



Impact & Benefits

Evidence for every forest decision.

FOREST TEAMS

Prioritise persistent loss alerts

Confirmed and provisional kept apart

RESTORATION TEAMS

Track canopy and biomass trends

Against a measured 2020 baseline

CARBON ANALYSTS

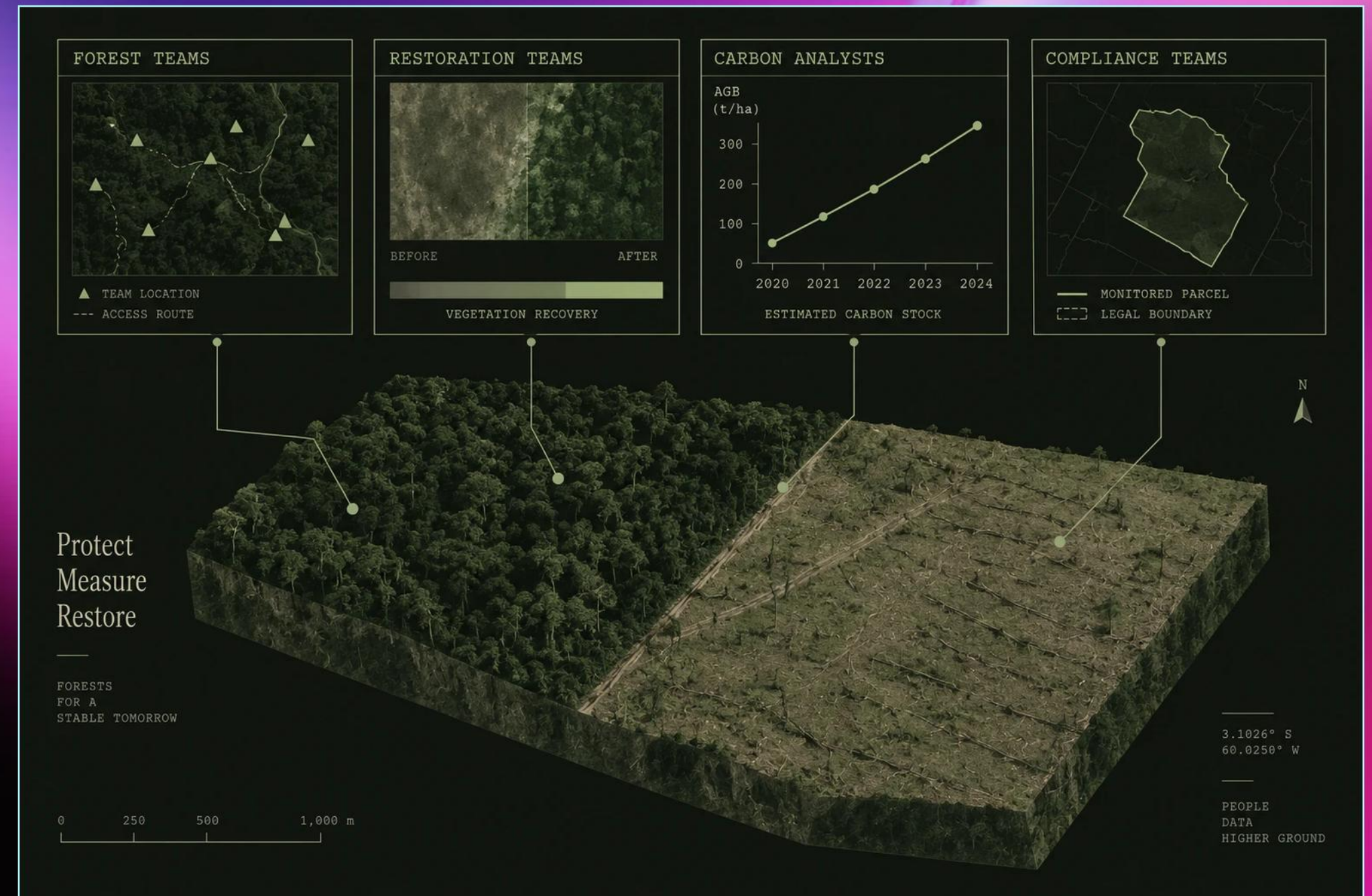
Trace stock change with its interval

Bias-adjusted area, not pixel counts

COMPLIANCE TEAMS

Evidence for EUDR, due Dec 2026

ESA WorldCover 2020 is the cut-off layer





Research & References

01 Official scope + submission rules

ORION PS-03 · Microsoft Club SIST

02 Satellite observations

Sentinel-1 processing

Sentinel-2 resolution

03 Biomass reference targets

NASA GEDI L4A · footprints + quality

04 Crown-scale data

NEON 0.1 m RGB

NEON 1 m CHM

05 Mask2Former + crown evidence

Cheng et al., 2022

Nakada et al., 2026 preprint

06 Crown segmentation benchmark

Detectree2 · Ball et al., 2023

07 Quantile biomass regression

XGBoost · official quantile objective

08 Time series + raster serving

Earth Engine CCDC

TiTiler

09 Biome labels

RESOLVE ecoregions · Dinerstein et al., 2017

10 Biomass-to-carbon formulation

IPCC 2006 · Forest Land · Table 4.3